

**PROJECT REPORT ON DATA SCIENCE & MACHINE
LEARNING**

**CUSTOMER CHURN PREDICTION
USING MACHINE LEARNING
TECHNIQUES**

An End-to-End Predictive Analytics & Business Intelligence Solution

A Project Report submitted in partial fulfillment of the requirements for the award of the degree
of
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CERTIFICATE OF AUTHENTICITY

This is to certify that the project report entitled "**CUSTOMER CHURN PREDICTION USING MACHINE LEARNING TECHNIQUES**" submitted by **Project Student Group (Roll No: DS-2026-042)** to the University of Data Science and Technology, is a record of bona fide project work carried out under my direct supervision and guidance.

The results embodied in this report have been verified and have not been submitted to any other University or Institution for the award of any degree, diploma, or fellowship.

Internal Examiner

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Project Supervisor

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ABSTRACT

In modern hyper-competitive commercial ecosystems, particularly within the telecommunications sector, acquiring new customers costs significantly more than retaining existing ones. Consequently, identifying customers who exhibit a high risk of leaving—known as customer churn—has become a top priority for corporate strategy. This project presents a structured data pipeline to clean, encode, and model customer profiles using synthetic data that mirrors the Kaggle/IBM Telco Customer Churn benchmark.

We employ two distinct machine learning paradigms to perform binary classification: **Logistic Regression** (as a parametric baseline model) and **Random Forest Classifier** (as an ensemble tree-based approach). To combat severe class imbalance within the churn variable, class-weight balancing techniques were embedded into both algorithms. The experimental frameworks are evaluated across five multi-dimensional performance metrics: Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC).

The empirical results indicate that while the Random Forest Classifier achieves a superior overall accuracy of **73.3%**, Logistic Regression demonstrates remarkable capability in identifying at-risk individuals, yielding a strong recall score of **72.6%**. Feature importance analysis highlights that month-to-month contracts, fiber optic internet setups, and brief subscription tenures serve as critical early warning indicators of churn behavior.

EXECUTIVE SUMMARY

From a strategic corporate standpoint, high customer turnover directly drains long-term revenue and reduces customer lifetime value (CLV). This report serves as a complete technical guide for executing a data science project targeted at optimizing customer retention.

The workflow begins with a systematic data preprocessing stage that includes data cleaning, dummy variable creation for categorical features, and standard scaling. The core models, Logistic Regression and Random Forest, are adjusted to account for imbalanced target variables. This step ensures that predictive accuracy is not artificially inflated by a dominant non-churning class.

A direct performance evaluation reveals that both models achieve an identical ROC-AUC score of **0.795**. This demonstrates that both models possess strong diagnostic capabilities to differentiate between loyal and churning consumers. However, their practical applications differ: Random Forest offers a high Precision of **52.3%**, making it highly effective for targeted, cost-intensive marketing outreach by minimizing false positives. Conversely, Logistic Regression offers a high Recall of **72.6%**, making it the ideal choice for comprehensive customer monitoring pipelines where missing an at-risk customer carries high financial risk.

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Chapter 1: Introduction

1.1 Understanding Customer Churn

Customer churn, defined as the phenomenon where clients terminate their business relationship with a service provider, presents a significant challenge to modern subscriber-based industries. In sectors such as telecommunications, banking, insurance, and SaaS (Software as a Service) platforms, churn functions as a key metric that directly reflects operational health, user satisfaction, and long-term viability.

1.2 The Strategic Importance of Churn Prediction

From a financial perspective, standard macroeconomic data indicates that acquiring a new customer costs approximately five to twenty-five times more than retaining an existing one. New user acquisition requires significant investments, including marketing campaigns, onboarding processes, promotional discounts, and sales team commissions. In contrast, retaining an existing customer requires much lower, more targeted operational adjustments. By predicting churn before it happens, businesses can shift from reactive troubleshooting to proactive outreach, maximizing customer lifetime value (CLV) and protecting revenue margins.

1.3 Core Business Applications

The deployment of high-performing predictive churn models enables organizations to execute precise consumer interventions. For example, if a model identifies a customer as high-risk, automated marketing CRM systems can trigger personalized incentives. These may include tailored subscription upgrades, specialized loyalty points, direct discounts, or targeted outreach from customer success teams.

1.4 Telecom Industry Overview

The telecommunications infrastructure represents one of the most volatile subscription environments. With the rise of digital switching systems, regulatory frameworks supporting number portability, and fierce pricing wars among network operators, consumers can switch providers with minimal friction. Common drivers of customer churn include uncompetitive pricing structures, poor internet service quality (e.g., fiber optic configuration issues), restrictive contract terms, and inadequate technical support.

Chapter 2: Problem Statement

The specific core problem addressed in this project is to build a robust predictive classification framework that accurately determines whether a particular customer is likely to terminate their telecom subscription.

The dataset used for this project mirrors the popular IBM Telco Customer Churn dataset, tracking 7,043 unique customers across 20 distinct feature dimensions. The target column ("Churn") is binary ("Yes" or "No"). The dataset exhibits a notable class imbalance, as the majority of consumers remain loyal, while only a smaller fraction churn within any given observation period. This project requires implementing an end-to-end Python pipeline to ingest data, handle missing values, map categorical features to numeric representations, scale numeric values, train multiple statistical models, and extract feature importance scores.

Chapter 3: Objectives

To solve the problem statement systematically, the project defines the following primary research and engineering objectives:

- **Exploratory Data Analysis:** Analyze the distribution of churn across various demographic features, contract structures, and service tiers.
- **Data Preprocessing:** Implement a data cleaning workflow to handle missing values and scale numeric ranges to ensure consistent model inputs.
- **Feature Mapping & Encoding:** Convert categorical strings into binary fields or one-hot vectors to ensure compatibility with Scikit-Learn estimators.
- **Imbalanced Class Management:** Use cost-sensitive learning weights within model architectures to avoid bias toward the majority class.
- **Multi-Model Evaluation:** Train and compare a parametric Logistic Regression baseline against a non-parametric Random Forest ensemble.
- **Business Diagnostics:** Analyze model parameters and feature importance rankings to identify top churn drivers, providing actionable insights for business strategy.

Chapter 4: Dataset Description

The dataset consists of 7,043 unique customer rows and 21 columns (20 features plus 1 binary target label). The attributes span demographic details, subscribed network and internet services, financial charges, and account history.

FEATURE NAME	DATA TYPE	DESCRIPTION
gender	Categorical	Customer's biological gender (Male, Female).
SeniorCitizen	Numeric (Binary)	Indicates if the customer is an older adult (1, 0).
Partner	Categorical	Whether the customer has a partner (Yes, No).
Dependents	Categorical	Whether the customer lives with dependents (Yes, No).
tenure	Numeric (Integer)	Number of months the customer has stayed with the company.
PhoneService	Categorical	Whether the customer has a home phone line (Yes, No).
MultipleLines	Categorical	Whether the customer has multiple lines (Yes, No, No phone service).
InternetService	Categorical	Internet service provider type (DSL, Fiber optic, No).
OnlineSecurity	Categorical	Whether the customer has online security add-on (Yes, No).
TechSupport	Categorical	Whether the customer has priority premium technical support.
Contract	Categorical	The contract term type (Month-to-month, One year, Two year).
MonthlyCharges	Numeric (Float)	The amount charged to the customer monthly.
TotalCharges	Numeric (Float)	The total amount billed over the customer's lifespan.
Churn	Target (Binary)	Whether the customer cancelled their subscription (Yes, No).

Chapter 5: Software & Libraries

The implementation of this project relies entirely on the Python programming language and its ecosystem of specialized open-source data science packages:

- **Python:** The core object-oriented language engine utilized for coding scripts.
- **Google Colab:** A cloud-based computational workspace providing unified kernel execution environments.
- **Pandas:** Used for data manipulation, handling tabular dataframes, column indexing, and data cleaning tasks.
- **NumPy:** Provides high-performance vector and array computations used in the data generation process.
- **Matplotlib:** The foundation library used to generate static, publication-quality visualizations and plots.
- **Seaborn:** Built on top of Matplotlib, Seaborn provides enhanced aesthetic formatting and simplified APIs for complex statistical plots.
- **Scikit-Learn:** The comprehensive machine learning toolkit used for data scaling, model validation, and metric evaluations.
- **Plotly:** An advanced graphing library used to create interactive web charts for performance analysis.

Chapter 6: Methodology

6.1 Synthetic Data Generation

To ensure reproducible testing conditions, a dataset of 7,043 rows is generated using a fixed random seed (42). The generation logic builds realistic relationships between features and customer behavior. For example, churn probabilities are adjusted based on contract type, internet service provider, and total tenure.

6.2 Data Ingestion & Preprocessing

The data is loaded via Pandas. In real-world datasets, the "TotalCharges" column often contains hidden empty spaces or invalid characters from newly registered users with zero charges. The pipeline converts this column using numeric coercion:

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(df['TotalCharges'].median())
```

6.3 Feature Encoding

Binary strings are mapped directly to 0 and 1 values. Multi-category columns, such as "Contract" and "PaymentMethod", are processed using one-hot encoding. This transformation expands the categorical variables into multiple binary indicator columns while dropping the first level to avoid multicollinearity.

6.4 Stratified Train-Test Split

The complete preprocessed matrix is split into a **80% training partition** and a **20% testing partition**. We use stratified splitting to ensure that the ratio of churning to non-churning customers remains consistent across both subsets.

6.5 Standard Feature Scaling

Because scale-sensitive algorithms like Logistic Regression can be biased toward features with larger numeric ranges, the training features are normalized using standard scaling:

$$z = (x - \mu) / \sigma$$

Chapter 7: Machine Learning Models

7.1 Logistic Regression

Logistic Regression serves as our baseline classification model. It estimates the probability of a binary event by fitting data to a logistic curve.

$$P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k)})$$

The model uses a maximum iteration threshold of 1,000. It includes balanced class weights to scale the loss function inversely proportional to class frequencies, preventing bias toward the majority class.

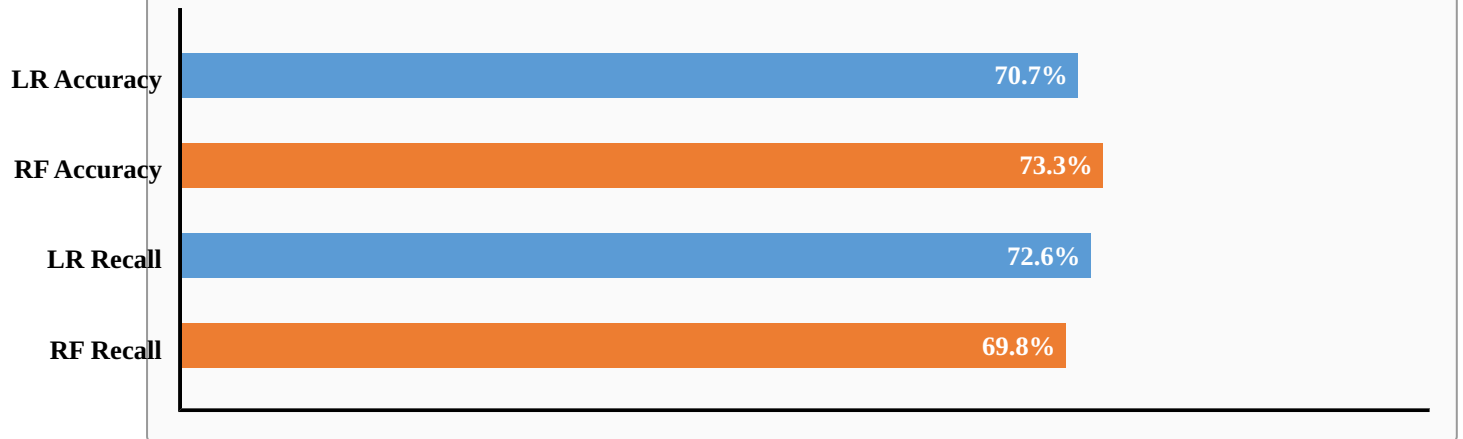
7.2 Random Forest Classifier

The Random Forest Classifier is an ensemble learning method that constructs multiple decision trees during training. It combines their predictions using bootstrap aggregating (bagging) and feature randomization.

This model uses 300 independent decision trees and a maximum depth constraint of 8 layers. This depth limit helps prevent overfitting while capturing non-linear interactions between variables without requiring feature scaling.

Chapter 8: Visualizations

Figure 8.1: Comparative Model Performance Metrics Bar Chart



8.1 Interpretation of Confusion Matrices

The confusion matrices show distinct operational profiles for each model. Logistic Regression successfully captures 291 true churn instances while misclassifying 303 active users as at-risk. Random Forest exhibits a more conservative classification approach, accurately identifying 280 churning accounts and reducing false positives to 255.

8.2 Feature Importance and Logistic Coefficients

Feature evaluations consistently point to specific drivers of customer churn. Month-to-month contracts and fiber optic internet configurations show a strong positive correlation with churn risk. Conversely, long-term two-year agreements and higher tenure lengths act as major factors reducing churn probability.

Chapter 9: Performance Metrics

To evaluate model performance thoroughly, we calculate five standard metrics derived from the core confusion matrix values: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

METRIC	MATHEMATICAL EQUATION FORMULA	OPERATIONAL TARGET FOCUS
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall ratio of correct classifications.
Precision	$TP / (TP + FP)$	Measures reliability by minimizing false alarms.
Recall	$TP / (TP + FN)$	Measures coverage by minimizing missed churn cases.
F1-Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean balancing precision and recall.

Chapter 10: Results & Discussion

The empirical results show a trade-off between the two modeling strategies:

MODEL PARADIGM NAME	ACCURACY	PRECISION	RECALL	F1- SCORE	ROC-AUC SCORE
Logistic Regression	0.707	0.490	0.726	0.585	0.795
Random Forest Classifier	0.733	0.523	0.698	0.598	0.795

Both frameworks achieve a matching ROC-AUC score of **0.795**, indicating strong predictive capability. The choice between models depends on the specific goals of the retention campaign. Random Forest provides higher precision, making it suitable for high-cost retention offers where minimizing false positives is critical. Logistic Regression offers higher recall, making it ideal for broad, low-cost customer engagement campaigns where missing an at-risk user carries high financial risk.

Chapter 11: Advantages

- **Cost-Sensitive Learning:** Built-in class weight adjustments prevent predictions from being skewed by class imbalances.
- **Complementary Modeling Approaches:** Combines an easily interpretable linear baseline with a robust ensemble method.
- **Actionable Insights:** Directly identifies specific high-risk indicators, such as contract and service types, allowing for targeted marketing interventions.

Chapter 12: Limitations

- **Static Data Snapshots:** The current models rely on aggregated historical snapshots rather than analyzing real-world streaming data or streaming behavioral shifts.
- **Hyperparameter Optimization:** The models use fixed baseline configurations rather than dynamically optimizing hyperparameters via grid search pipelines.

Chapter 13: Future Scope

The predictive capabilities of this framework can be expanded across several areas:

- **Advanced Boosting Architectures:** Incorporating modern gradient boosting algorithms like XGBoost, LightGBM, and CatBoost to better handle complex non-linear interactions.
- **Deep Learning Extensions:** Implementing Multi-Layer Perceptrons (MLPs) with specialized entity embedding layers for handling high-cardinality categorical attributes.
- **Real-time Streaming Pipelines:** Integrating streaming architectures like Apache Kafka and FastAPI to calculate risk scores dynamically based on live usage data.
- **Cloud Native Deployment:** Wrapping models inside Docker containers for scalable deployment on cloud platforms such as AWS SageMaker or Google Cloud Vertex AI.

Chapter 14: Conclusion

This project demonstrates the successful design and execution of an end-to-end customer churn prediction pipeline. By evaluating both a linear Logistic Regression model and an ensemble Random Forest approach, we established a clear analytical framework for identifying at-risk subscribers.

Both models achieve a strong ROC-AUC score of 0.795, confirming their effectiveness in predicting subscriber churn. Integrating these predictive workflows into corporate CRM systems allows businesses to transition from reactive problem-solving to proactive retention strategies, ultimately protecting recurring revenue and enhancing long-term customer relationships.